TEMASEK POLYTECHNIC
SCHOOL OF INFORMATICS & IT

DIPLOMA IN APPLIED ARTICIFICAL INTELLIGENCE
DIPLOMA IN BIG DATA & ANALYTICS
COMMON ICT PROGRAMME
DIPLOMA IN CYBERSECURITY & DIGITAL FORENSICS
DIPLOMA IN IMMERSIVE MEDIA & GAME DEVELOPMENT
DIPLOMA IN INFORMATION TECHNOLOGY

AY2025/2026 OCTOBER SEMESTER

PROJECT (40%)

NETWORK AND CLOUD TECHNOLOGY (CCF1C04)

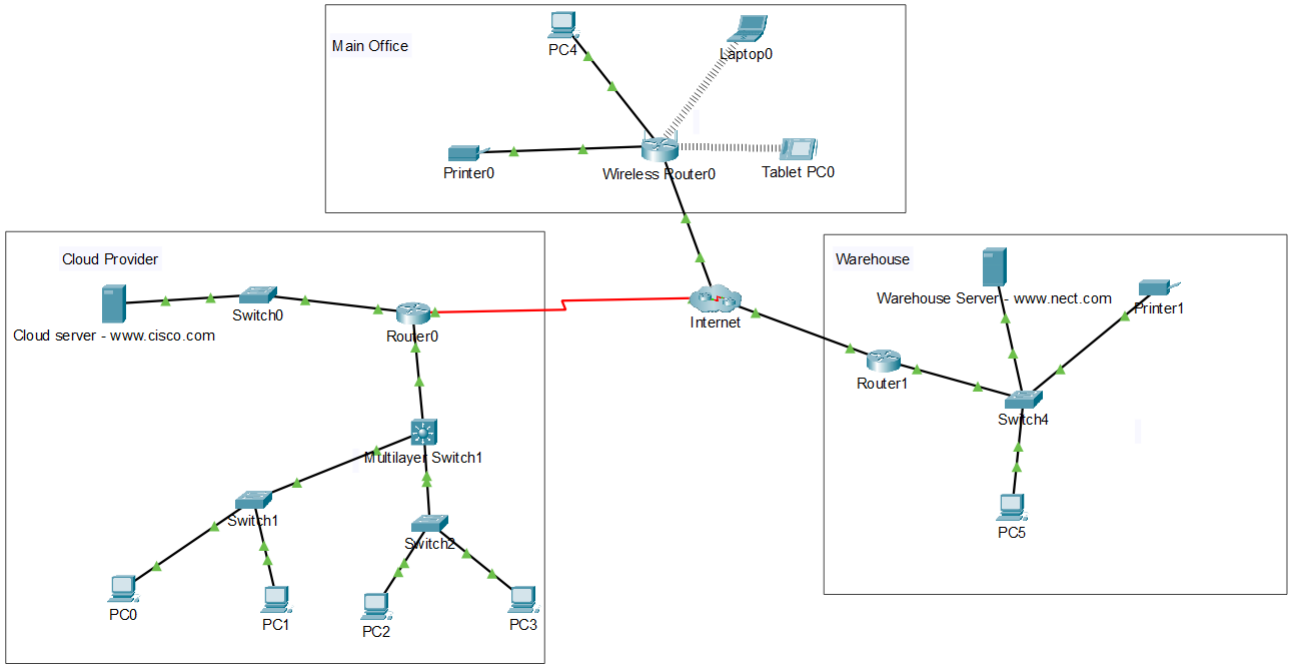
SUBJECT LEVEL: 1

INSTRUCTIONS TO CANDIDATES

1. This project consists of **7** pages (including cover page and appendix).
2. The objective is to assess students on the followings.
   a. Knowledge and skills in implementing networking services.
3. This project is to be done individually.
4. Students would be assessed individually for the project. Each student is required to submit.

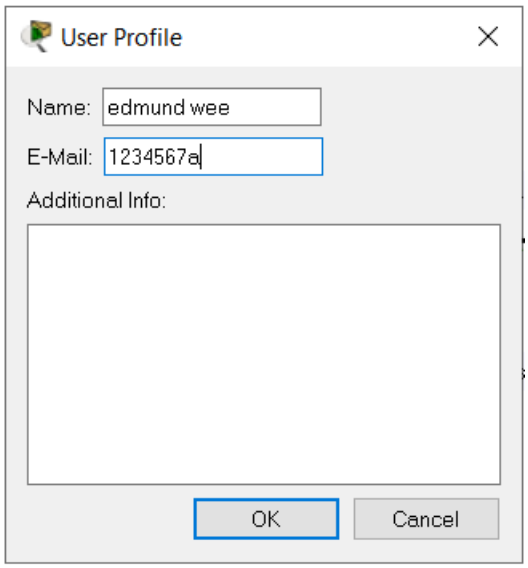
Scenario

You are to configure the NECT Pte Ltd's Main Office and Warehouse using the project Packet Tracer file provided. Configure the project Packet Tracer file so that it looks like the below network topology.

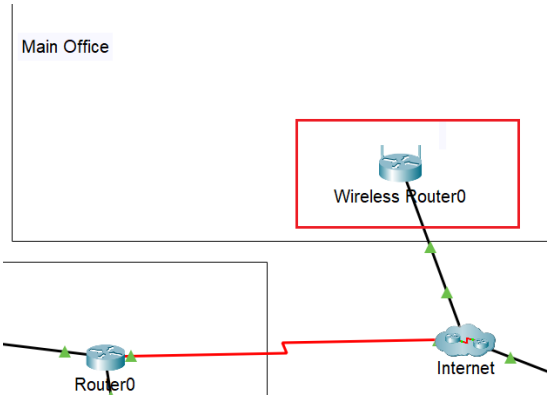


Additional Info:

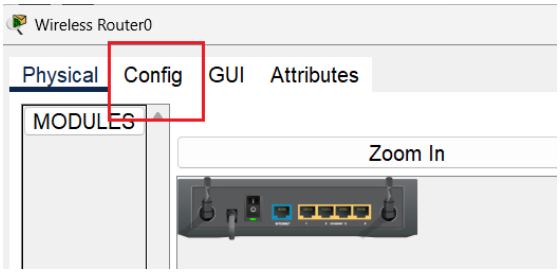
- Download the ***Project.pka*** Packet Tracer activity file found in POLITEMall. Please enter your name and admin number (e.g. see screenshot below) and click OK before attempting to complete the tasks.



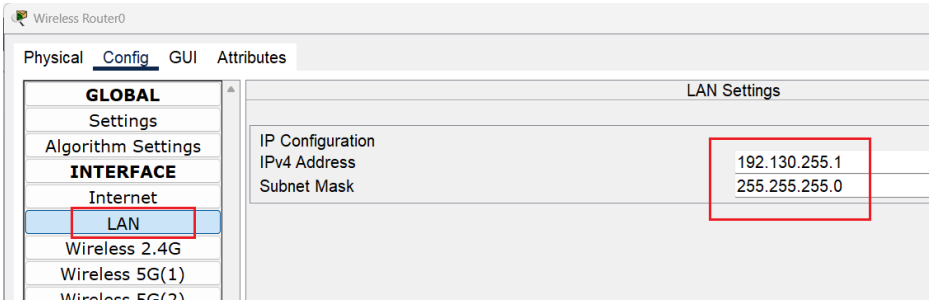
- Cloud Provider web server IP Address: 8.8.8.8
- Cloud Provider web server name: www.cisco.com
- ISP DNS: 203.116.1.78
- Main Office IP Subnet: 192.130.x.0/24 where x is your class index number
    - Main Office Gateway: 192.130.x.1 where x is your class index number
    - Please follow the below steps to change the Main Office wifi router Gateway IP address
        1. Click on the Main Office wifi router



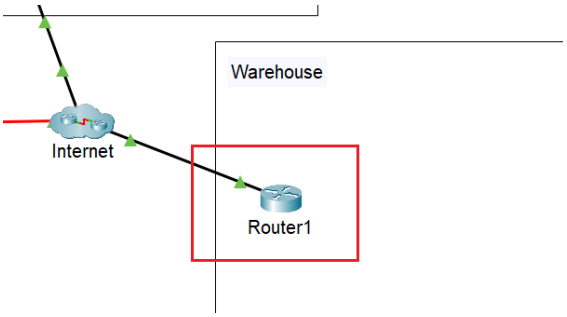
        2. Select the config tab



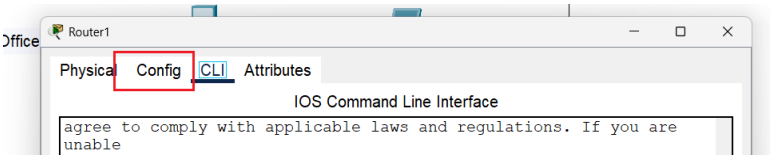
        3. Change values in the LAN according to your IP address



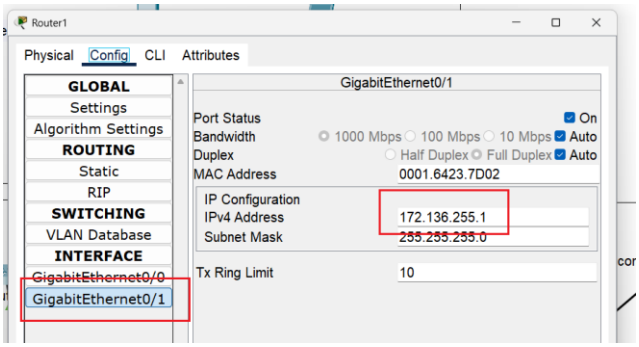
- Warehouse IP subnet: 172.136.x.0/24 where x is your class index number
    - Warehouse Gateway: 172.136.x.1 where x is your class index number
    - Please follow the below steps to change the Warehouse Router Gateway IP address
        1. Click on the Warehouse router



2. Select the config tab



3. Change the GigabitEthernet0/1 according to your IP address



Implement the following tasks in the provided project packet tracer file (to be completed individually)

1. Add the networking devices and end devices as shown in the given network topology
2. Implement wifi for the Main Office
    a. SSID: NECT (For laptop to connect)
        i. Security mode: WPA2
        ii. Encryption AES
        iii. Passphrase: ILoveNECT
    b. SSID: Guest (For Tablet to connect)
        i. Security mode: Disabled

3. Implement DHCP for both Main Office's wifi router and Warehouse server respectively
    a. Start IP address: y.y.x.50
    b. Maximum number of users: 100
    * y being the IP subnet given for the Main Office and Warehouse

4. Implement the followings on the Warehouse server

a. IP address: 172.136.x.10
b. HTTPS with the following message in the homepage



c. DNS and add a DNS record so that warehouse users can access the Warehouse Server via www.nect.com

5. Implement the following test cases:
    a. Main Office
        i. Users (i.e. PC, laptop and tablet) in Main Office able to:
            1. Get IP address via DHCP
            2. Access Cloud Provider web server by using both it's IP address and name
            3. All end device in the Main Office should be able to ping to the Cloud Provider web server's IP address
        ii. Printer:
            1. Set IP address 192.130.x.11
    b. Warehouse
        i. Users (i.e. PC) in Warehouse able to:
            1. Get IP address via DHCP
            2. Access Warehouse server via www.nect.com
            3. All end device in the Warehouse should be able to ping to the Cloud Provider web server's IP address
        ii. Printer:
            1. Set IP address 172.136.x.11

Submission

You are required to submit your individual Packet Tracer pka and report via POLITEMall.

Refer to the Teaching Plan for the submission deadline.

Assessment Scheme

Your project will be assessed based on the following criteria.

- Tasks (15%)
- Presentation of implemented tasks (15%)

- Questions and Answers (10%)
    - Questions asked will be what you have learnt for this module

Note: Plagiarism is not allowed and will result in a zero for the assignment; including AI generated work e.g., ChatGPT. You are required to cite any sources used in your project.

Late submissions

Late submissions will be imposed as follows.

- Late and < 1 day – 10% deduction from absolute marks given for the part of the work e.g. if the assignment was worth 100 marks, you were given 75 marks for the work, after the penalty you are left with 65 marks i.e. 75 – 100x10%
- Late >= 1 and < 2 days – 20% deduction from absolute marks
- Late >= 2 days – no marks will be awarded

~~ End of Project ~~

Appendix A - Report Template

SCHOOL OF INFORMATICS & IT

Diploma in [ ]

Project Report

Student Name (Matric Number):

Tutorial Group :

Tutor :

Submission Date :

Declaration of Originality

I am the originator of this work and I have appropriately acknowledged all other original sources used as my references for this work.

I understand that Plagiarism is the act of taking and using the whole or any part of another person's work, including work generated by AI, and presenting it as my own.

I understand that Plagiarism is an academic offence

and if I am found to have committed or abetted the offence of plagiarism in relation to this submitted work, disciplinary action will be enforced.

Table of Content

1. Introduction

2. Implementation of Tasks
   2.1 Tasks #1 (Provide detailed steps you have taken to complete the tasks, use screenshots where applicable)
   2.2 Tasks #2
   2.3 Tasks #3
   2.4 Tasks #4
   2.5 Tasks #5

3. Any additional relevant documentation
   …

4. Conclusion

Appendixes

A- References
B- Screenshots of ping results
C- …